F18CE2 MV Calculus and Real Analysis B: Tutorial 9

1 Let $f : [a, b] \to \mathbb{R}$ be a bounded function and let $P = [x_0, x_1, x_2, \ldots, x_n]$ be a partition of [a,b]. Define the upper and lower sums $U(P, f)$ and $L(P, f)$.

2 Compute the upper sum $U(P, f)$ of the function $f(x) = 9(x-1)^2$ for the partition $P = \{0, 2/3, 4/3, 2\}$.

3 Let $f : [0, 3] \to \mathbb{R}$ be a bounded function. Let P and Q be partitions of $[0, 3]$ given by $P = \{0, 3\}$ and $Q = \{0, 2, 3\}$. Prove in detail that $U(P, f) \geqslant U(Q, f)$.

4 Let f be the function defined on $[0, 1]$ by $f(x) = 5$ when $x = \frac{1}{2}$ and $f(x) = 0$ otherwise. Compute $U(P_n, f)$ and $L(P_n, f)$ for the partition $P_n = \{0, 1/2 - 1/n, 1/2 + 1/n, 1\}, n > 2$. Deduce that f is integrable on $[0, 1]$ and find $\int_0^1 f(x)\, dx$.

5 Let $g : [0, 10] \to \mathbb{R}$ be defined by

$$g(x) = \begin{cases} -2\,, & \text{for } 0 \leqslant x < 6, \\ 5\,, & \text{for } x = 6, \\ 4\,, & \text{for } 6 < x \leqslant 10. \end{cases}$$

Let P_n be the partition given by $P_n = \{0, 6 - 1/n, 6 + 1/n, 10\}$.

Find $U(P_n, g)$ and $L(P_n, g)$ and hence show that g is integrable on $[0, 10]$ and find $\int_0^{10} g(x)\, dx$.

6 Let $f : [a, b] \to \mathbb{R}$ be increasing on $[a, b]$. Compute the upper and lower sums $U(P, f)$ and $L(P, f)$ for the partition $P = \{x_0, x_1, x_2, \ldots, x_n\}$.

7 Let $f : [a, b] \to \mathbb{R}$ be integrable on $[a, b]$, and $f(x) \geqslant I(f)$ for all $x \in [a, b]$. Prove that for any partition P of $[a, b]$, the lower Riemann sum satisfies

$$L(P, f) \geqslant I(f)\,(b - a).$$

Deduce that

$$\int_a^b f(x)\, dx \;\geqslant\; I(f)\,(b - a)\,.$$

By considering $f(x) = x^{-1}$, prove that for $a > 0$, $\ln(1 + a^{-1}) \;\geqslant\; (a + 1)^{-1}$.

8 For the following functions, evaluate the integral $\displaystyle\int_0^1 f$ by using Riemann sums, using the partitions:

$$P_n = \left\{0, \frac{1}{n}, \frac{2}{n}, \frac{3}{n} \ldots, \frac{n-1}{n}, 1\right\}, \qquad n = 2, 3, \ldots.$$

Also, work out these integrals using the fundamental theorem of calculus.

(a) $f(x) = x$;

(b) $f(x) = x^2$. Hint: $\displaystyle\sum_{r=0}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$.

9 (a) Consider the function $g : [-1, 1] \to \mathbb{R}$ defined by

$$g(x) := \begin{cases} 0, & -1 \leqslant x < 0, \\ 1, & 0 \leqslant x < 1. \end{cases}$$

Calculate the values of the function $G : [-1, 1] \to \mathbb{R}$ defined by

$$G(x) := \int_{-1}^{x} g(t)\, dt, \quad x \in [-1, 1],$$

and show that G is continuous on $[-1, 1]$, but is not differentiable on $(-1, 1)$.

(b) Now consider the function $g : [-1, 1] \to \mathbb{R}$ defined by

$$g(x) := \begin{cases} 0, & x \neq 0, \\ 1, & x = 0. \end{cases}$$

Show that in this case G is differentiable on $(-1, 1)$, but $G' \neq g$.

These examples show that the continuity conditions in the Fundamental Theorem of Calculus are necessary.